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Roll No.....

AU-502 (GS)
B.E. V Semester Examination, June 2020
Grading System (GS)
Operation Research and Supply Chain
Time : Three Hours

Maximum Marks : 70

Note: i) Answer any five questions.
ii) All questions carry equal marks.

1. What do you understand by Bull-whip effect? What are its causes and how it can be controlled?
2. Explain the following terms
 - i) Basic feasible solution
 - ii) Degenerate solution.
3. State the common and distinguishing features of the assignment and transportation models?
4. Compare Single server model (M/M/1) and Multiple server models (M/M/S).
5. Solve graphically the following LPP.
Maximize $Z = 8x_1 + 16x_2$
Subject to $x_1 + x_2 \leq 200$
 $x_2 \leq 125$
 $3x_1 + 6x_2 \leq 900$
Where $x_1, x_2 \geq 0$
6. How will you control the inventories of a manufacturing organization? Discuss the various inventory costs associated with this organisation.
OR
Explain solution of LP by two variables.
7. Explain VED analysis used in inventory control. Write advantages and limitations.
OR
How information sharing and strategic partnership is done in SCM? Explain.
8. Answer any two of the following :
 - a) Discuss the ABC analysis of inventory system.
 - b) Describe the used of information technology is supply chain.
 - c) How a assignment problem is a special case of transportation problem.
 - d) How heuristic and metaheuristic algorithm can be differentiated.
